

CS471: Introduction to Artificial Intelligence

Assignment 5

In recent years, distributed energy resources such as solar and wind power stations present havoc to the distribution portion of the grid. The intermittent fluctuations of power coming from the generation source creates power quality issues and instability challenges, which make the possibility of system failures more frequent and causes extensive damage to the power system infrastructure. To tackle this issue, optimizing wind turbine performance is essential, and the analysis of current wave data is crucial. Accurate identification of oscillatory signals within this data is challenging but vital, as these signals indicate operational concerns.

In this assignment, you will work with this wind farm dataset.

Datasets are found in: 'train.csv' and 'testing.csv'

The starter code for the assignment is in main.py

There are two classes, one being a normal class and the other being an oscillation class. The goal is to perform classification using any machine learning algorithm of your choice. You can use the Scikit-learn toolbox to develop your models.

1. Describe what is a binary classification and multi-class classification problem and specify if the current problem is a binary or a multi-class one. (1 point)
2. Perform pre-processing if necessary. (2 points)
3. Fit a machine learning algorithm of your choice on the training dataset. (1 point)
Explain the reason behind selection of machine learning algorithm (2 points)
4. Tune hyperparameters (if any) based on the performance on the validation set or using cross-validation. Plot the hyperparameter value on the x-axis and performance metric on the y-axis. (2 points)
5. Evaluate it on test data and generate a classification report (2 points)

CS471 Assignment #5 Rubric	
Criteria	Pts
Describe what is a binary classification and multi-class classification problem and specify if the current problem is a binary or a multi-class one.	/ 1 pts
Perform pre-processing if necessary.	/ 2 pts
Fit a machine learning algorithm of your choice on the training dataset. Explain the reason behind selection of machine learning algorithm.	/ 3 pts
Tune hyperparameters (if any) based on the performance on the validation set or using cross-validation. Plot the hyperparameter value on the x-axis and performance metric on the y-axis.	/ 2 pts
Evaluate it on test data and generate a classification report	/ 2 pts
	/ 10 pts